UF Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lecture 5A - Visual Sensory System I

EIN3241: Human Factors and Ergonomics
Spring 2026

February 20, 2026

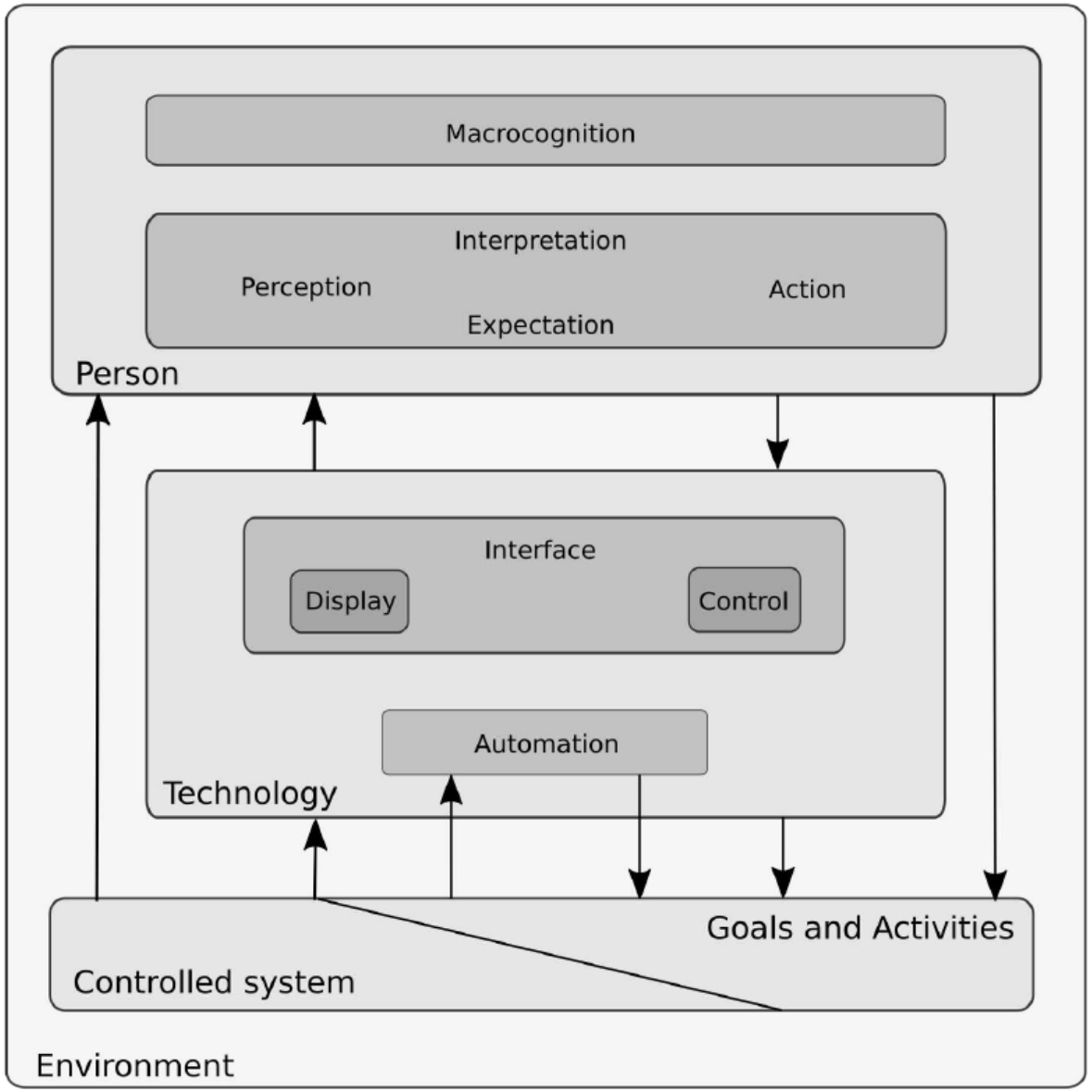
Dr. Jad A. Atweh

Admin Issues

- Homework 3 due Friday.
- Lab 3 (Strength) due Wednesday.
- Last Quiz before the Midterm
  - Quiz 4: Friday, February 27
- Midterm in 17 days!
  - Monday, March 9
  - 8:20 – 10:10 PM in FLG270
  - Up until HIP

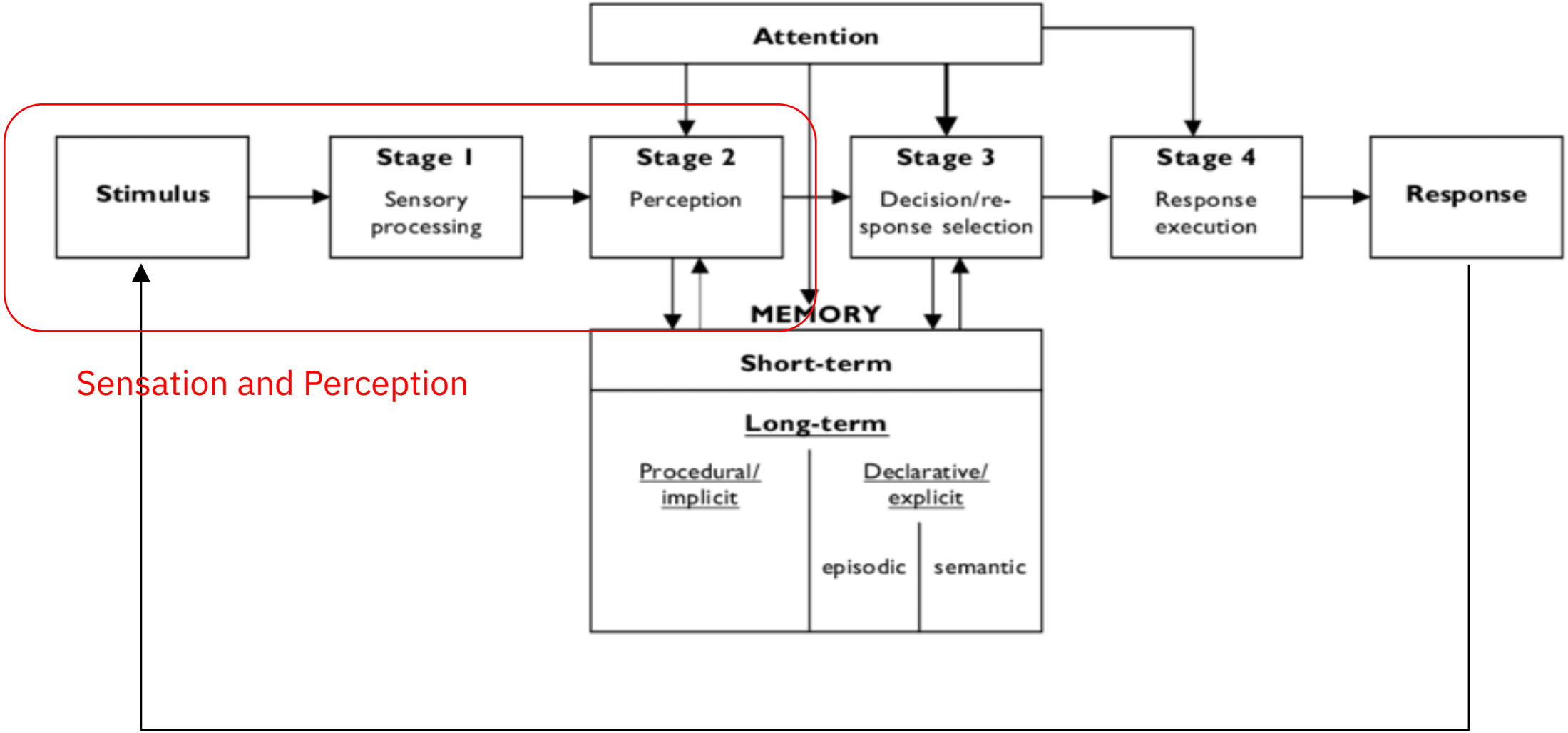
Why are we talking about this?

- Sensation and perception are the first stages of the human cognitive process
- Human perception is often non-linear
- The systems view of human-interactive systems.
- About 95-99% of information is channeled through the eyes or ears…



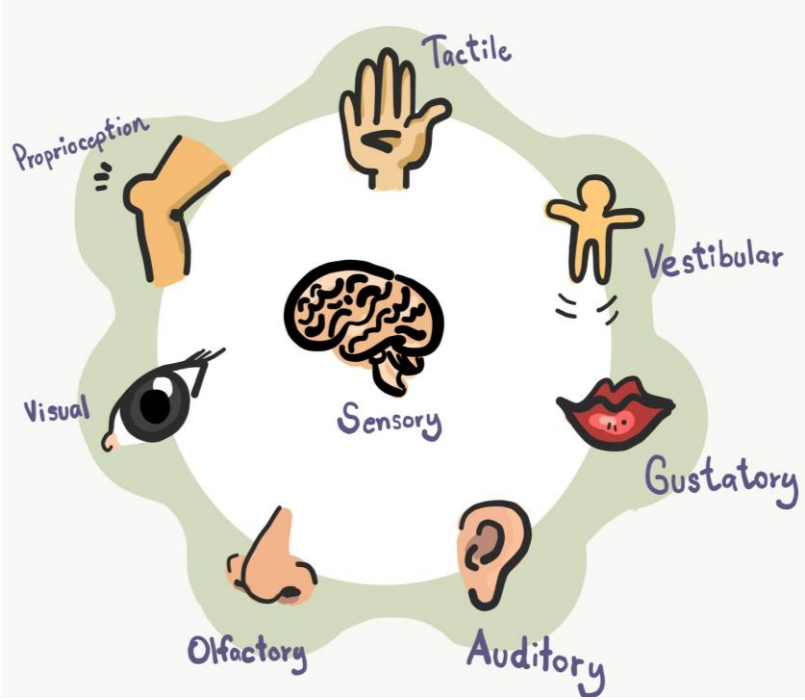
The information process model of human cognition

- We will dive deeper into each box throughout the class…



Sense, Sensation, and Perception

- **Sense** – physical system that receives physical stimulation from surrounding environment and translates that stimulation into an electrochemical message
- **Sensation** – detecting physical energy in the environment and encoding it as neural signals. Neurons transmit (and processes) the information from the sense organ to the brain.
- **Perception** – processing of information done by the brain – mental processes that organize and interpret sensory information that has been transmitted to the brain.



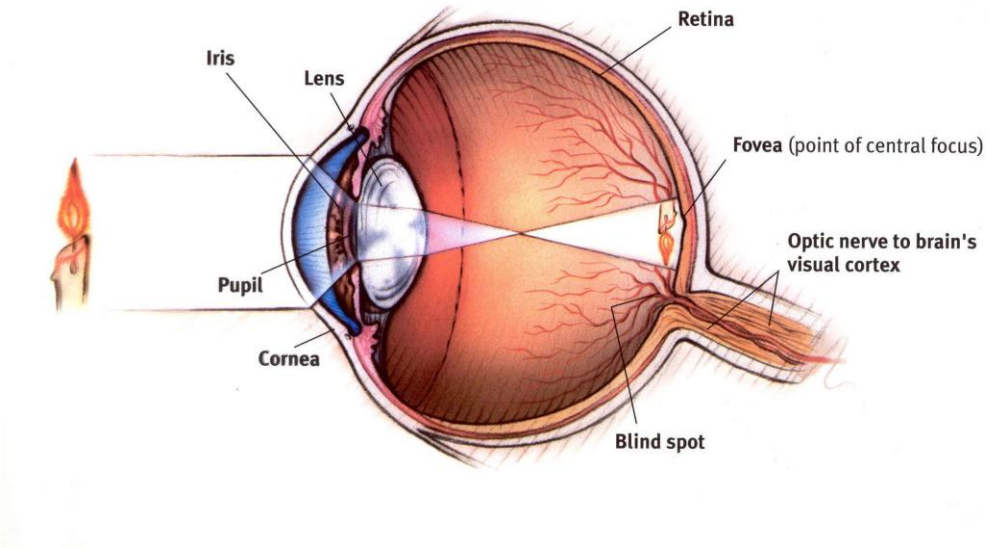
Sensory Systems

- What are the properties of the stimuli (light, acoustic waves, pressure, vibrations, etc.) and how do we measure it?
- What are the sensory organs (eyes, ears, skin)?
  - What are the properties of the receptors and how does it affect our perception?
    - Adaptation (Dark adaptation, temporary threshold shift, touch adaptation, etc.)
    - Depth perception
    - Haptic perception
- What are the important measures of our perception?
  - Acuity, contrasts, loudness
- What are important HF implications?
  - Design (improve perception, detection, and recognition)
  - Safety (protect our sensory systems, or protect the human)

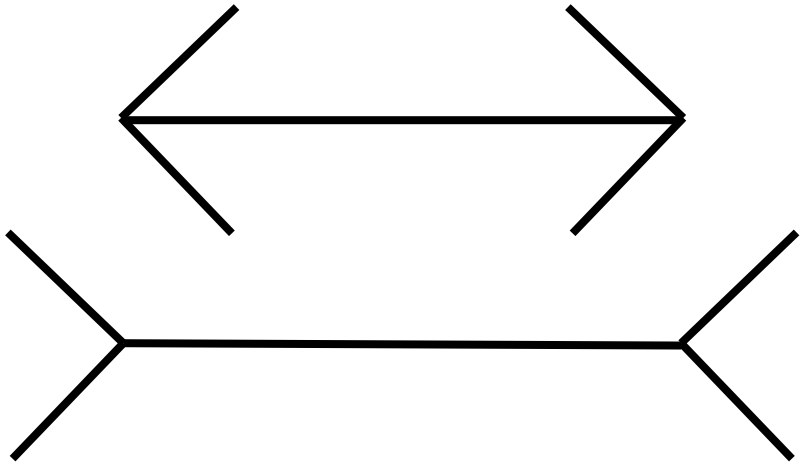
Visual sensory system

- Demand vs. capacity:
  - Information reaching human from environment = more than 100,000 bits/s
  - Information processing capacity of human operator = less than <u>20</u> bits/s
    - Bit = Logarithmic transformation of number of states of sensory signal (e.g., signal with 2 states = $Log_2(2) = 1\ bit$)
    - 20 bits is ~1.05M signals states/s.
- Creates high potential for translational errors and misunderstanding

Visual sensory system



- System functions based on detection of light
- Functions of system answer two questions:
  - What is it? (perception/recognition)
  - Where is it? (location)
- Optical illusions/paradoxes can occur:
  - Components of object can distort perceptual processing of overall form (feature extraction).



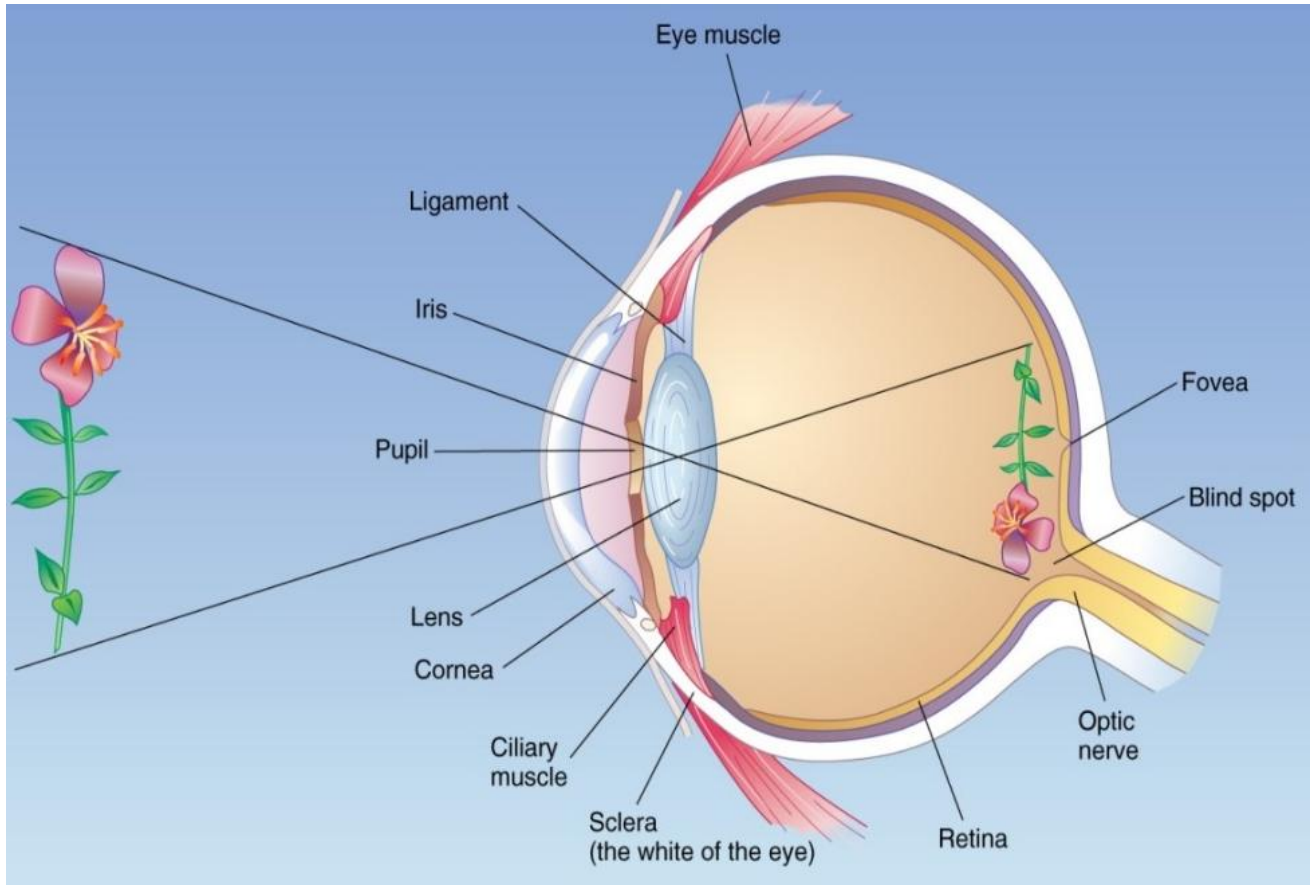
Enclosed line lengths are same.

Many competing theories why this illusion occurs.

  - Visual perception cannot always be trusted!

Anatomy of the Eye

- Visual sensory organ
- Structures:
  - Cornea: Protective outer surface over lens.
  - Pupil: Aperture of which size is controlled by iris:
    - Dilates or constricts ("adaptation" to dark and bright conditions)
    - Also changes in response to other physiological changes
    - Can change 2-8 mm in diameter
  - Lens: Changes in shape through ciliary muscle action to focus objects (or "**accommodate**")

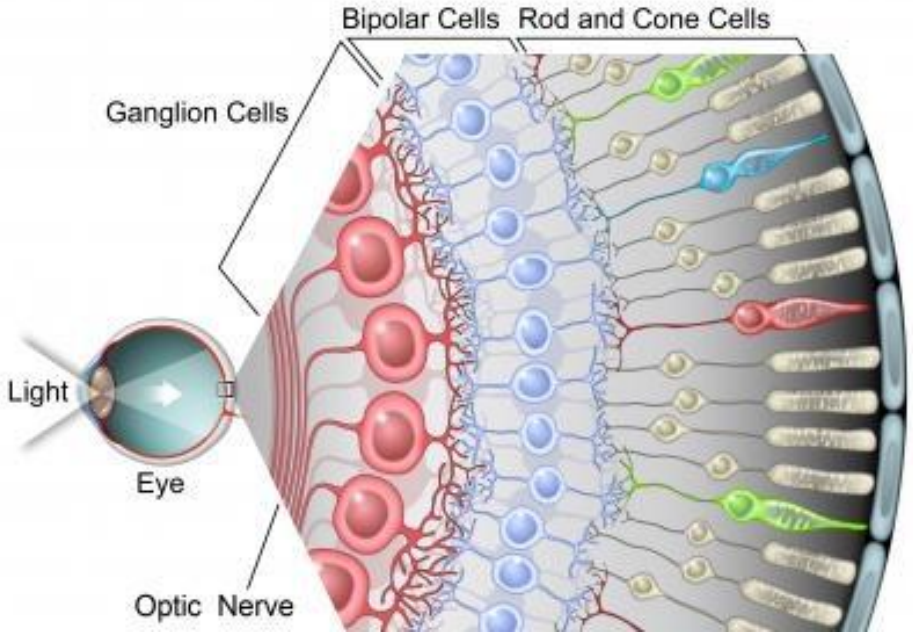


Visual images are inverted as they pass through lens.

Anatomy of the Eye

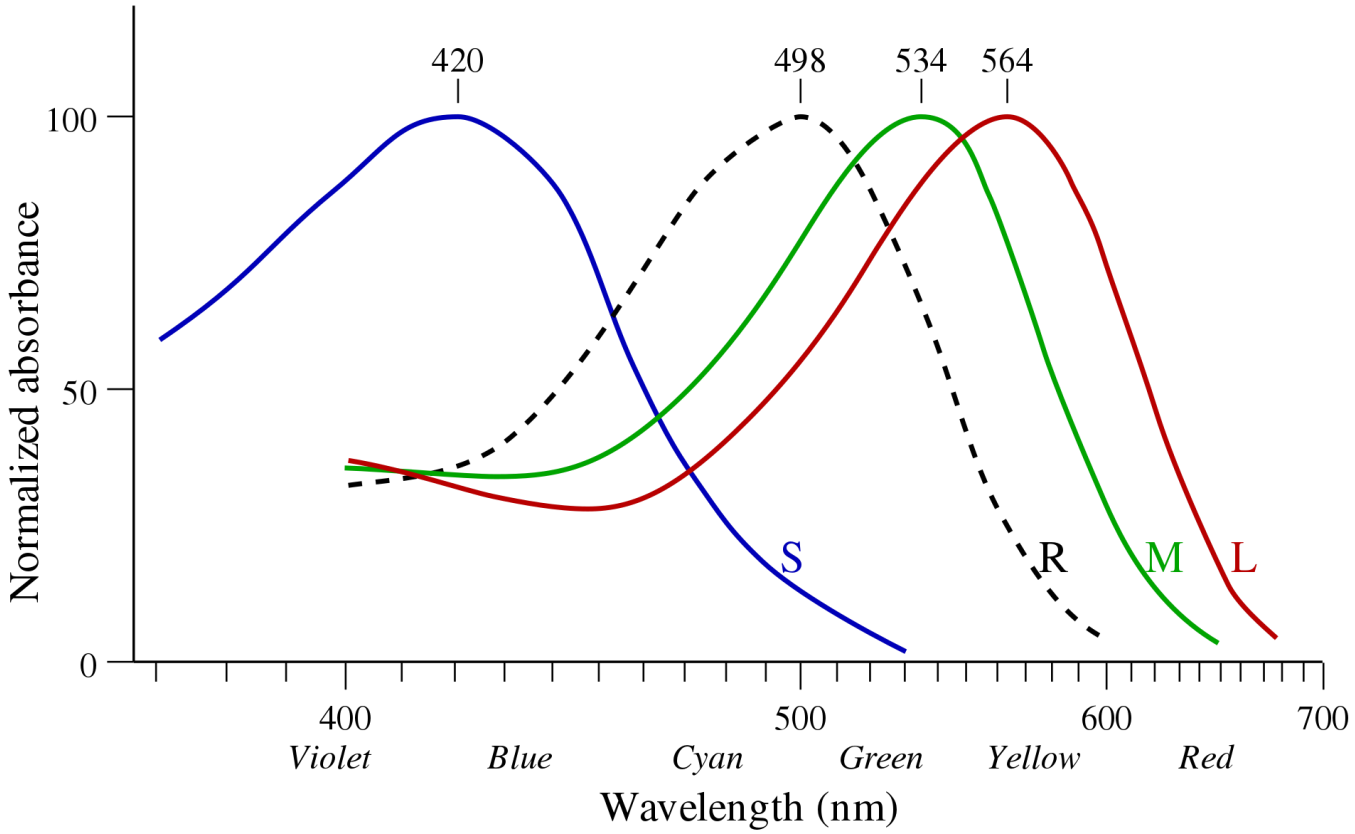
- Retina converts electromagnetic waves to electrochemical signals
- Three types of photoreceptor cells

1. Photosensitive retinal ganglion cells:
   - Discovered in the 1990s
   - Control pupil dilation, circadian rhythms
2. Rod cells:
   - Adapt to lower light levels (e.g., nighttime (scotopic) vision)
   - High sensitivity but poor acuity
     - Difficult to discriminate colors at night – vision is limited to light vs. dark
     - Difficult to discriminate fine detail at night – vision is limited to peripheral (motion) vs. foveal (reading)

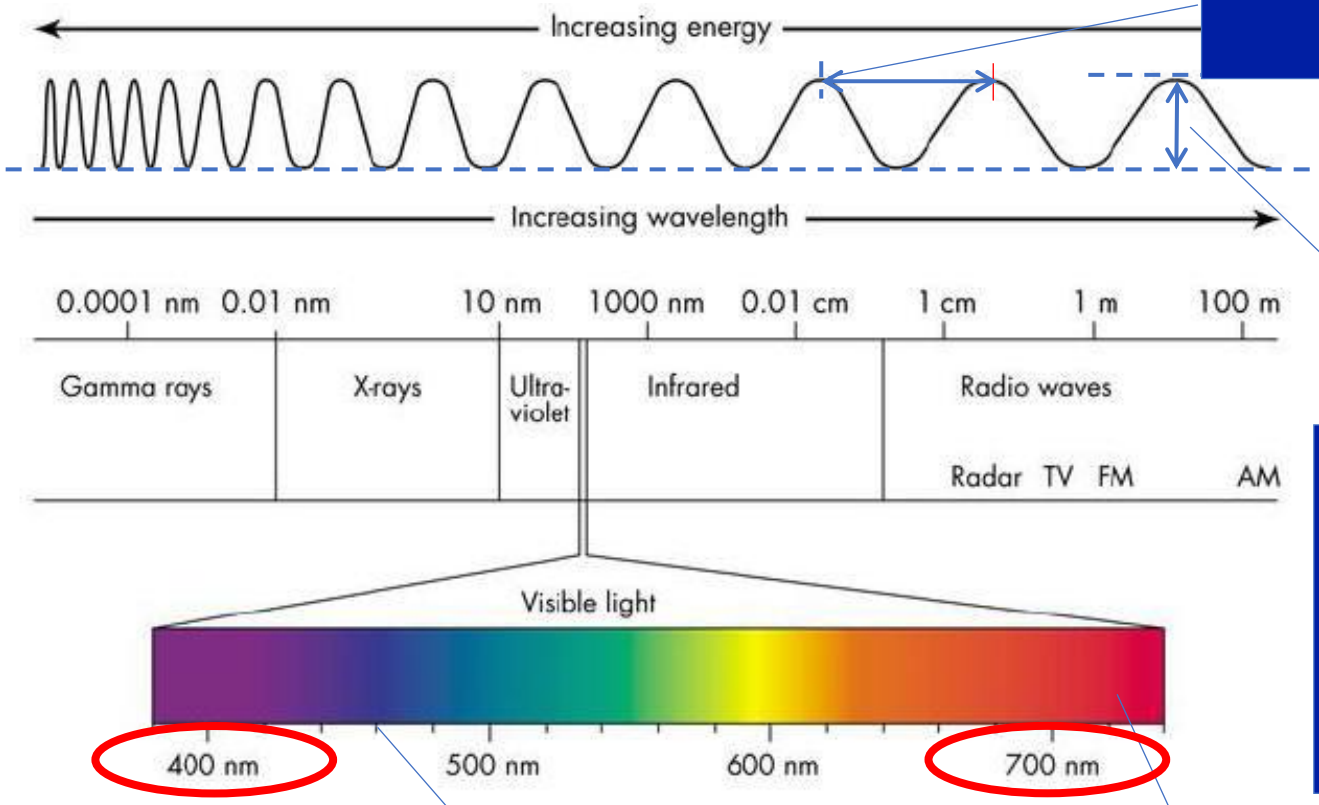


Anatomy of the Eye

3. Cone cells:
   - Three types of cone cells: Short, Medium, Long
   - Active at higher light levels (e.g., daytime (photopic) and color vision)
   - High acuity but low sensitivity
     - Concentrated ONLY in fovea
     - Discriminate color and support fine detail
     - Less sensitive to changes in illuminance



Visual spectrum



Wavelength (perceived as hue or specific tone of color)

Amplitude (perceived as brightness)

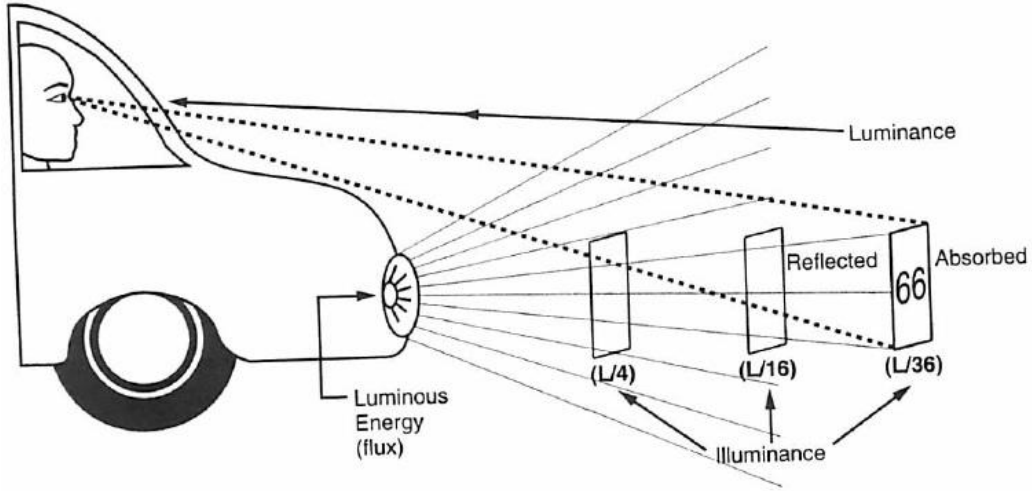
Human eye is most sensitive to green wavelength. Important display information (like numbers on instruments) are generally green (or sometimes yellowish green) in color.

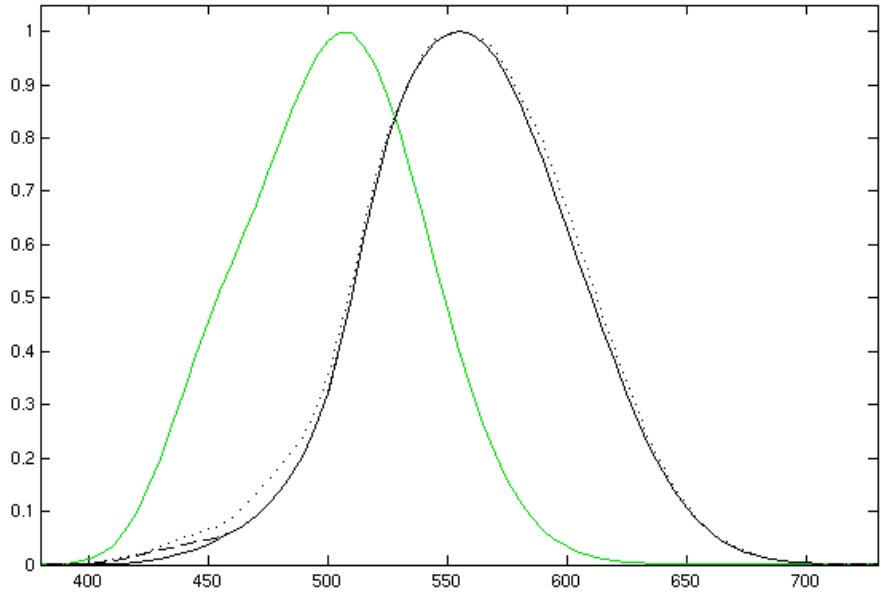
Blue color:
Higher in energy and shorter wavelength

Red color:
Lower in energy and longer wavelength

Physical measures of light

- Luminous flux (actual light energy of source)
  - Power radiated over wavelengths sensitive to human eye
  - Unit: lumen (lm)
- Illuminance (amount of light hitting object you see)
  - Depends on distance between object and light source
  - Declines with square distance
  - Unit: lux (lx), foot candles (fc)
- Luminance (perceived brightness of given object)
  - Amount of light leaving surface in particular direction
  - Depends on space in which object is perceived
  - Unit: candela per square meter (candela/m^2), foot lambert (fl)





$$\text{Reflectance (\%)} = \frac{luminance\ (fl)}{illuminance\ (fc)}$$

Lighting Standards

Examples of Recommended and Naturally Occurring Illuminances

Area/Activity	Illuminance on a Horizontal Surface (lx)
Clear sky in summer	150,000
Overcast sky in summer	16,000
Performance of extremely low contrast tasks. Certain surgical operations	10,000
Textile inspection	1,500
Office work (pencil handwriting, poorly reproduced documents)	1,000
Precise assembly work	1,000
Drawing offices, meat inspection, chain stores	750
Office work (without VDUs)	500
Office work (data entry)	500
Office work (VDU conversational tasks)	300
Heavy engineering	300
Rough assembly work	200
Minimum illuminance for manual handling tasks (NIOSH)	150
Rarely visited places where little perception of detail is required (e.g., railway platforms)	50
Public spaces in otherwise unlit areas	20
Good street lighting	10
Emergency lighting[a]	2
Moonlight	0.5

Factors Affecting Vision

- Vision is affected by:
  - Internal factors (or personal factors)
  - External factors (the design of the surrounding environment)

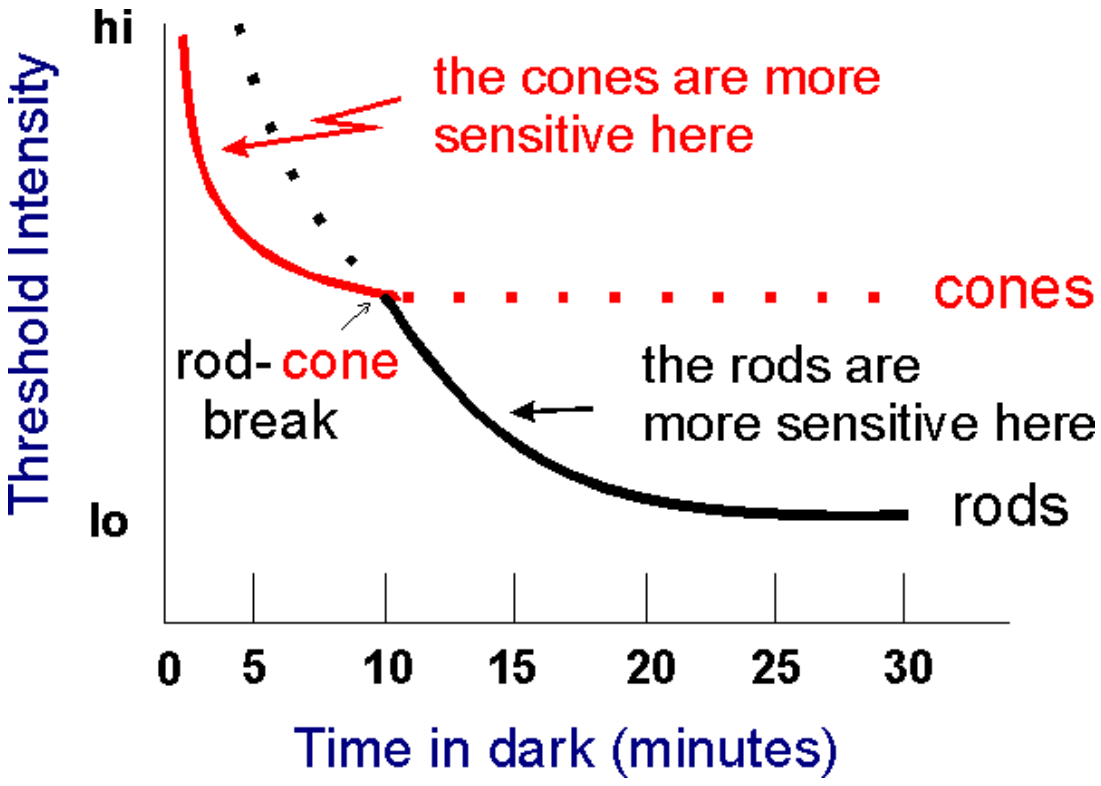
http://usd-apps.usd.edu/coglab/IntroPeripheral.html

Internal Factors Affecting Vision

- Internal factors relate to human physiological characteristics
  1. Dark adaptation
  2. Phorias
  3. Age
  4. Visual acuity
  5. Near point accommodation
  6. Peripheral vision
  7. Normal line of sight (NLS)
  8. Color discrimination
  9. Depth perception

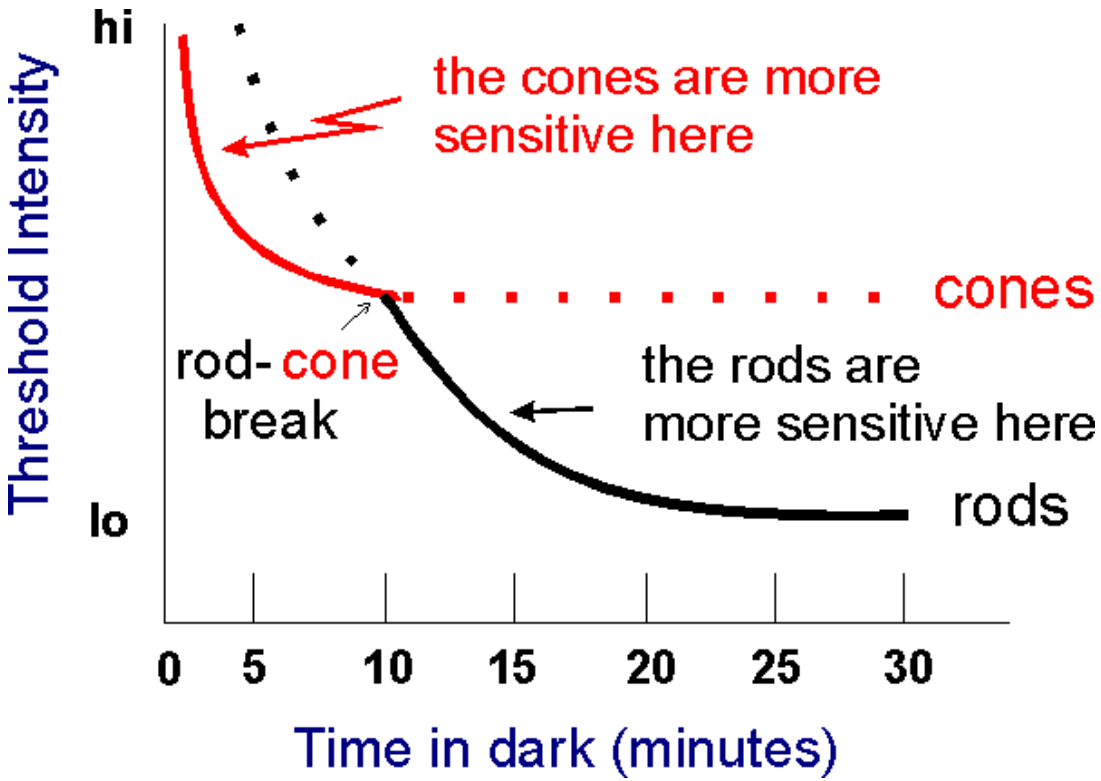
1. Vision adaptation (dark and light)

- Adaptation - Ability to see under different lighting conditions
- Going from light to dark
  - Body suppresses cone threshold for active night vision
  - Rods threshold also decreases – Allows detection of small changes in illuminance and motion
  - Takes about 30 min to dark adapt, (but just a few minutes to light adapt)
  - Sensitivity to contrast among surfaces increases with dark adaptation



1. Vision adaptation (dark and light)

- 0-10 min in dark - Cones require less light to reach threshold response (perception of stimulus) than rods
- 10+ min in dark - rods require less light for perception of stimulus.
- Point at which rods become more sensitive is called "rod-cone break".
- Glare can temporarily destroy rod cell sensitivity to low spatial frequencies (e.g., fail to detect dim road hazards due to water reflectance from headlights)



Human Factors Application

- Why are red lights used in cockpits?
  - Rods and cones are NOT equally sensitive to all wavelengths
  - Rods are insensitive to longer λ (wavelength)

- Red light (low energy, long λ) least detrimental to pilot night vision
  - Rod cells are relatively insensitive to red light (receptors do not fatigue under red light); as if pilot had already spent time in dark
  - Red light minimally triggers pupil constriction so pilot can continue to see dim lights outside of cockpit (e.g., other planes)

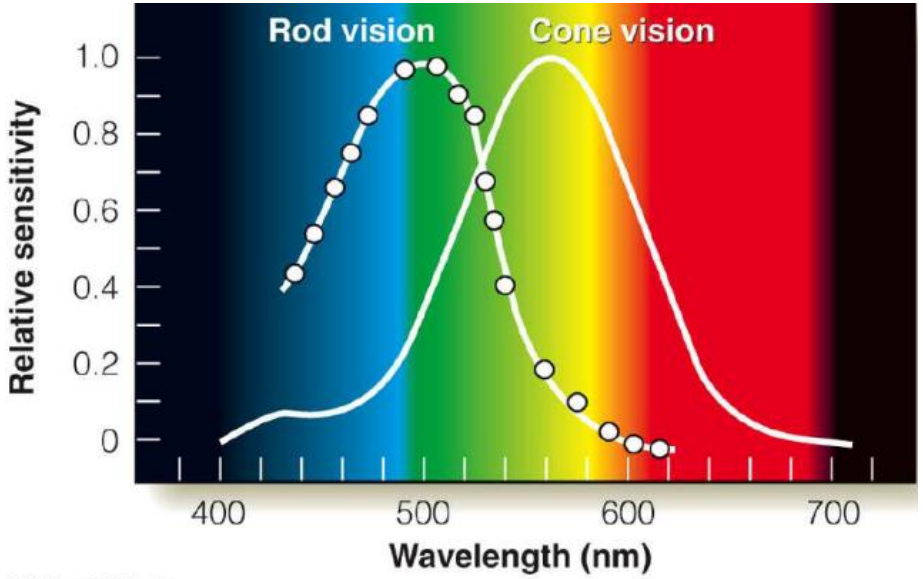


Figure Spectral sensitivity curves for rod vision (left) and cone vision (right).

Human Factors Application

- Two problems with red light in cockpit:
  1. Human eye is <u>less sensitive</u> to appearance of red light than any other color of spectrum = poor reaction times
  2. Pilots cannot see red print (used on navigation charts) under red light:
     - Cockpits have small, narrow-beam white light, used sparingly
     - Pilots are instructed to close one eye when using white light sources to preserve night adaptation of eye.

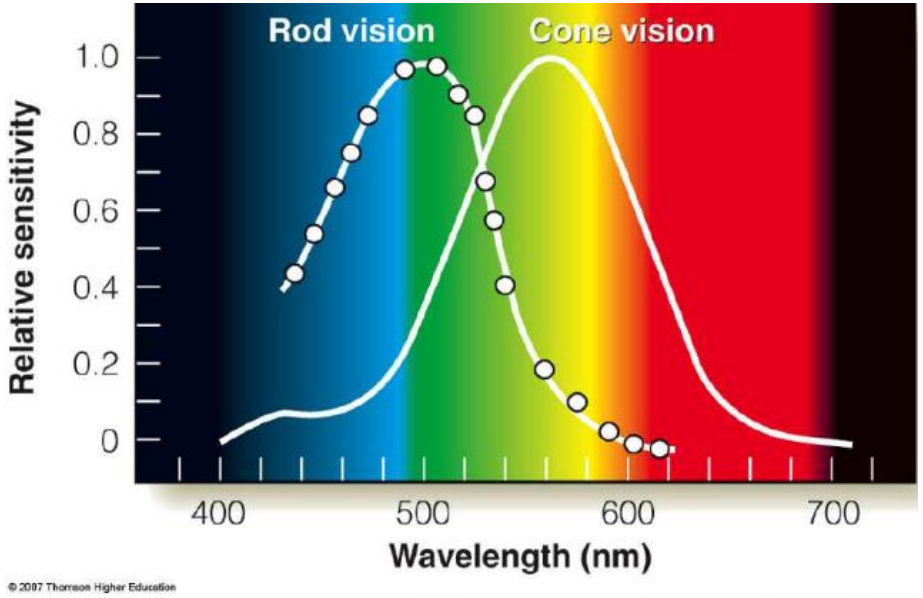


Figure Spectral sensitivity curves for rod vision (left) and cone vision (right).

2. Phorias

- Some people eyes converge too much, while others tend not to converge enough. Such conditions are called **phorias**.
- People with phorias also see one image but experience notable stress to the eye muscles, making them more vulnerable to rapidly developing visual fatigue.

3. Age

- With age, ciliary muscles weaken and visual performance declines.
- A young adult normally reads from a distance of about 30 cm (1 ft). At 60, the near point of accommodation is around 83cm (2.75 ft).
- In industry, reading requirements, including character size, may be based on the limiting age of the expected user population.

UF
Herbert Wertheim
College of Engineering
UNIVERSITY of FLORIDA
LEADING THE CHARGE, CHARGING AHEAD